Grade 10 Term 1 Lesson 1 Practice Activity 2
Revenue, Costs, Profit, and Payoff Tables

Contents

Evaluated criteria

C1 Characterises decision-making problems under uncertainty by identifying decision alternatives, states of nature, consequences, and the economic or financial context of the decision.

C3 Calculates profit as revenue minus costs across states of nature and organises the resulting payoffs in payoff tables.

1 Introduction

Problems:

A decision problem under uncertainty requires a choice of an *alternative* A_i before an uncertain event reveals a *state of nature* S_j. The alternative is a row of the table and the state is a column. Their combination produces a *consequence*; its monetary *payoff* is profit $P(A_i, S_j)$. *Revenue* $R(A_i, S_j)$ is income earned, *cost* $C(A_i, S_j)$ is the value of resources used, and

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

A payoff table places every $P(A_i, S_j)$ at row A_i and column S_j. The objective here is to construct that complete table, not to choose the best alternative.

Revenue and cost are independent axes. Their general models are

$$R(A_i,S_j)=q(A_i,S_j)\left[r^A(A_i)+r^S(S_j)\right]+F^{R,A}(A_i)+F^{R,S}(S_j),$$
$$C(A_i,S_j)=q(A_i,S_j)\left[c^A(A_i)+c^S(S_j)\right]+F^{C,A}(A_i)+F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned}P(A_i,S_j)=q(A_i,S_j)\left[r^A(A_i)+r^S(S_j)\right]+F^{R,A}(A_i)+F^{R,S}(S_j)\\ -\left\{q(A_i,S_j)\left[c^A(A_i)+c^S(S_j)\right]+F^{C,A}(A_i)+F^{C,S}(S_j)\right\}.\end{aligned}$$

Here a *variable component* is quantity $q(A_i,S_j)$ multiplied by a per-unit rate; a *fixed component* is a lump sum independent of quantity. The superscript A marks an alternative-based component whose argument is A_i, and the superscript S marks a state-based component whose argument is S_j. The quantity $q(A_i,S_j)$ depends on both arguments and may change in every cell. Thus, inspect the arguments: A_i means that the value depends on the alternative, S_j means that it depends on the state, and (A_i,S_j) means that it may depend on both. A fixed lump sum is not multiplied by quantity. A component absent from one model is zero. A directly supplied $R(A_i,S_j)$ or $C(A_i,S_j)$ needs no decomposition.

Reusable procedure.

1. Identify the alternatives, uncertain event, states, consequences, and units.
2. Separate all revenue data from all cost data.
3. Classify each item as variable or fixed and as alternative-based or state-based; treat absent components as zero.
4. For every cell, calculate $R(A_i,S_j)$ from its revenue model and $C(A_i,S_j)$ independently from its cost model.
5. Subtract revenue minus costs:
$$P(A_i,S_j)=R(A_i,S_j)-C(A_i,S_j).$$
6. Put alternatives in rows and states in columns; check every cell and all units.

Paired ladder overview. AV, SV, AF, and SF denote alternative-variable, state-variable, alternative-fixed, and state-fixed components. Within each tier, the cost list is a deliberate permutation of the revenue list; matching structures are not assumed.

Tier	Revenue structures, in problem order	Cost structures, in problem order
Direct	direct	direct
One component	AV; SV; AF; SF	SV; AF; SF; AV
Two components	AV+SV; AV+AF; AV+SF; SV+AF; SV+SF; AF+SF	AV+SF; SV+AF; SV+SF; AF+SF; AV+SV; AV+AF
Three components	AV+SV+AF; AV+SV+SF; AV+AF+SF; SV+AF+SF	AV+SV+SF; AV+AF+SF; SV+AF+SF; AV+SV+AF
Complete	AV+SV+AF+SF	AV+SV+AF+SF

1.1 Introductory General Problem (C1–C3)

A school enterprise chooses A_1 or A_2 before demand is S_1 or S_2. The consequence is profit in thousand USD. Total revenues are

$$R(A_1,S_1)=20,\quad R(A_1,S_2)=13,$$
$$R(A_2,S_1)=18,\quad R(A_2,S_2)=15.$$

The directly supplied total costs are

$$C(A_1, S_1) = 9, \quad C(A_1, S_2) = 8,$$
$$C(A_2, S_1) = 10, \quad C(A_2, S_2) = 9.$$

Questions/Tasks.

C1 Interpret the decision elements.

C3 Build the payoff table.

C1 - Interpreting the decision.

Element	Description	
Alternatives	**A1:** A_1	**A2:** A_2
Statesofnature	**S1:** S_1	**S2:** S_2
Events	Demand revealed after the enterprise choice	
Consequences	Profit in thousand USD for each alternative–state pair	

C3 - Calculating revenue, cost, and payoff.

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	20	9	$20 - 9 = 11$
(A_1, S_2)	13	8	$13 - 8 = 5$
(A_2, S_1)	18	10	$18 - 10 = 8$
(A_2, S_2)	15	9	$15 - 9 = 6$

Alternative	S_1	S_2
A_1	11	5
A_2	8	6

1.2 Advanced General Problem (C1–C3)

A community business chooses A_1 or A_2 before demand is S_1 or S_2. The quantities are

$$q(A_1, S_1) = 2, \quad q(A_1, S_2) = 1,$$
$$q(A_2, S_1) = 3, \quad q(A_2, S_2) = 2.$$

In thousand USD, the alternative-based revenue rates are

$$r^A(A_1) = 10, \qquad r^A(A_2) = 11,$$

and the state-based revenue rates are

$$r^S(S_1) = 5, \qquad r^S(S_2) = 3.$$

The alternative-based fixed revenues are

$$F^{R,A}(A_1) = 7, \qquad F^{R,A}(A_2) = 9,$$

and the state-based fixed revenues are

$$F^{R,S}(S_1) = 6, \qquad F^{R,S}(S_2) = 7.$$

The alternative-based cost rates are

$$c^A(A_1) = 3, \qquad c^A(A_2) = 4,$$

and the state-based cost rates are

$$c^S(S_1) = 1, \qquad c^S(S_2) = 2.$$

The alternative-based fixed costs are

$$F^{C,A}(A_1) = 8, \qquad F^{C,A}(A_2) = 10,$$

and the state-based fixed costs are

$$F^{C,S}(S_1) = 5, \qquad F^{C,S}(S_2) = 3.$$

Questions/Tasks.

C1 Interpret the decision elements.

C3 Build the payoff table.

C1 - Interpreting the decision.

Element	Description	
Alternatives	**A1:** A_1	**A2:** A_2
Statesofnature	**S1:** S_1	**S2:** S_2
Events	Demand revealed after the business choice	
Consequences	Profit in thousand USD from the complete revenue and cost models	

For every alternative–state pair, revenue and cost are calculated independently:

$$\begin{aligned}
R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j),\\
C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j),\\
P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j).
\end{aligned}$$

C3 - Calculating revenue, cost, and payoff.

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$2(10+5)+7+6=43$	$2(3+1)+8+5=21$	$43-21=22$
(A_1, S_2)	$1(10+3)+7+7=27$	$1(3+2)+8+3=16$	$27-16=11$
(A_2, S_1)	$3(11+5)+9+6=63$	$3(4+1)+10+5=30$	$63-30=33$
(A_2, S_2)	$2(11+3)+9+7=44$	$2(4+2)+10+3=25$	$44-25=19$

Alternative	S_1	S_2
A_1	22	11
A_2	33	19

2 Direct Revenue and Cost Models

Problems:

In a direct model, the total revenue $R(A_i, S_j)$ and total cost $C(A_i, S_j)$ are supplied for every alternative–state pair. For each cell, the payoff is calculated directly as

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

2.1 Local Café Investment (C1–C3)

A small café is deciding whether to launch a new line of artisan desserts. The transaction is daily café sales, where revenue is earned from sales and costs include the resources required to operate each menu option. The owner must choose between:

1. A_1: **Launch the dessert line**
2. A_2: **Keep the current menu**

The possible states of nature are:

1. S_1: High traffic
2. S_2: Low traffic

The directly estimated revenues and costs, in thousand USD, are:

1. Launch dessert line:
   i) High traffic: $R(A_1, S_1) = 58$, $C(A_1, S_1) = 16$
   ii) Low traffic: $R(A_1, S_2) = 27$, $C(A_1, S_2) = 19$
2. Keep current menu:
   i) High traffic: $R(A_2, S_1) = 49$, $C(A_2, S_1) = 21$
   ii) Low traffic: $R(A_2, S_2) = 41$, $C(A_2, S_2) = 21$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Launch dessert line	**A2:** Keep current menu
Statesofnature	**S1:** High traffic	**S2:** Low traffic
Events	Actual foot traffic after the decision	
Consequences	Profit in thousand USD for each alternative–state pair	

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	58	16	$58 - 16 = 42$
(A_1, S_2)	27	19	$27 - 19 = 8$
(A_2, S_1)	49	21	$49 - 21 = 28$
(A_2, S_2)	41	21	$41 - 21 = 20$

The resulting payoff table is:

Alternative	High traffic	Low traffic
Launch dessert line	42	8
Keep current menu	28	20

2.2 Green Energy Production Mix (C1–C3)

A renewable-energy company must select an electricity-generation mix. The transaction is annual electricity sales, where revenue is earned from electricity generated and sold, while costs arise from operating the selected generation system.

The company is comparing three alternatives:

1. A_1: Build wind turbines
2. A_2: Build solar farms
3. A_3: Build a balanced hybrid system

Three weather states are possible:

1. S_1: Windy
2. S_2: Sunny
3. S_3: Cloudy

The directly estimated revenues and costs, in thousand USD, are:

1. Wind turbines:
   i) Windy: $R(A_1, S_1) = 132$, $C(A_1, S_1) = 42$
   ii) Sunny: $R(A_1, S_2) = 78$, $C(A_1, S_2) = 33$
   iii) Cloudy: $R(A_1, S_3) = 62$, $C(A_1, S_3) = 32$
2. Solar farms:
   i) Windy: $R(A_2, S_1) = 77$, $C(A_2, S_1) = 42$
   ii) Sunny: $R(A_2, S_2) = 143$, $C(A_2, S_2) = 48$
   iii) Cloudy: $R(A_2, S_3) = 63$, $C(A_2, S_3) = 38$
3. Hybrid system:
   i) Windy: $R(A_3, S_1) = 112$, $C(A_3, S_1) = 42$
   ii) Sunny: $R(A_3, S_2) = 107$, $C(A_3, S_2) = 42$
   iii) Cloudy: $R(A_3, S_3) = 96$, $C(A_3, S_3) = 36$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Wind turbines	**A2:** Solar farms	**A3:** Hybrid system
Statesofnature	**S1:** Windy	**S2:** Sunny	**S3:** Cloudy
Events	Weather outcome affecting electricity generation		
Consequences	Profit in thousand USD for each strategy–weather combination		

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	132	42	$132 - 42 = 90$
(A_1, S_2)	78	33	$78 - 33 = 45$
(A_1, S_3)	62	32	$62 - 32 = 30$
(A_2, S_1)	77	42	$77 - 42 = 35$
(A_2, S_2)	143	48	$143 - 48 = 95$
(A_2, S_3)	63	38	$63 - 38 = 25$
(A_3, S_1)	112	42	$112 - 42 = 70$
(A_3, S_2)	107	42	$107 - 42 = 65$
(A_3, S_3)	96	36	$96 - 36 = 60$

The resulting payoff table is:

Strategy	Windy	Sunny	Cloudy
Wind turbines	90	45	30
Solar farms	35	95	25
Hybrid system	70	65	60

3 One-Component Models

Problems:

In a one-component model, revenue contains exactly one nonzero component and cost contains exactly one nonzero component. Each component may be variable or fixed and may depend on the alternative or the state. The four paired patterns in this section are:

Problem	Revenue model	Cost model
5	$R(A_i, S_j) = q(A_i, S_j)r^A(A_i)$	$C(A_i, S_j) = q(A_i, S_j)c^S(S_j)$
6	$R(A_i, S_j) = q(A_i, S_j)r^S(S_j)$	$C(A_i, S_j) = F^{C,A}(A_i)$
7	$R(A_i, S_j) = F^{R,A}(A_i)$	$C(A_i, S_j) = F^{C,S}(S_j)$
8	$R(A_i, S_j) = F^{R,S}(S_j)$	$C(A_i, S_j) = q(A_i, S_j)c^A(A_i)$

Thus, the four one-component structures—AV, SV, AF, and SF—appear exactly once as a revenue structure and exactly once as a cost structure.

3.1 Custom Printing Product Line (C1–C3)

A custom-printing cooperative must select the product line that it will produce before market conditions are known. The transaction is the sale of production lots. Revenue per lot depends on the selected product, while cost per lot depends on the market state.

The cooperative is comparing two alternatives:

1. A_1: Produce premium notebooks
2. A_2: Produce canvas bags

The possible states of nature are:

1. S_1: Strong demand and stable material prices
2. S_2: Weak demand and higher material prices

The quantities, measured in production lots, are

$$q(A_1, S_1) = 6, \quad q(A_1, S_2) = 4,$$
$$q(A_2, S_1) = 5, \quad q(A_2, S_2) = 3.$$

The alternative-based revenue rates, in thousand USD per lot, are

$$r^A(A_1) = 12, \qquad r^A(A_2) = 15.$$

The state-based cost rates, in thousand USD per lot, are

$$c^S(S_1) = 5, \qquad c^S(S_2) = 8.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Produce premium notebooks	**A2:** Produce canvas bags
Statesofnature	**S1:** Strong demand and stable material prices	**S2:** Weak demand and higher material prices
Events	Market conditions revealed after the product-line decision	
Consequences	Profit in thousand USD for each product-line–market-state pair	

The only revenue component is an alternative-based variable component:

$$R(A_i, S_j) = q(A_i, S_j) r^A(A_i).$$

The only cost component is a state-based variable component:

$$C(A_i, S_j) = q(A_i, S_j) c^S(S_j).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j) r^A(A_i) - q(A_i, S_j) c^S(S_j).$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate revenue and cost independently:

$$R(A_i, S_j) = q(A_i, S_j) r^A(A_i),$$
$$C(A_i, S_j) = q(A_i, S_j) c^S(S_j),$$
$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$6(12) = 72$	$6(5) = 30$	$72 - 30 = 42$
(A_1, S_2)	$4(12) = 48$	$4(8) = 32$	$48 - 32 = 16$
(A_2, S_1)	$5(15) = 75$	$5(5) = 25$	$75 - 25 = 50$
(A_2, S_2)	$3(15) = 45$	$3(8) = 24$	$45 - 24 = 21$

The resulting payoff table is:

Alternative	Strong demand and stable prices	Weak demand and higher prices
Premium notebooks	42	16
Canvas bags	50	21

3.2 Event Equipment Fleet (C1–C3)

An event-equipment provider must choose the size of the fleet that it will prepare before rental demand is known. Revenue per rental contract depends on the demand state, while the preparation cost is a fixed amount determined by the selected fleet.

The provider is comparing two alternatives:

1. A_1: Prepare a compact fleet
2. A_2: Prepare an expanded fleet

The possible states of nature are:

1. S_1: High rental demand
2. S_2: Low rental demand

The numbers of rental contracts are

$$q(A_1, S_1) = 6, \quad q(A_1, S_2) = 3,$$
$$q(A_2, S_1) = 8, \quad q(A_2, S_2) = 4.$$

The state-based revenue rates, in thousand USD per contract, are

$$r^S(S_1) = 9, \qquad r^S(S_2) = 6.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 22, \qquad F^{C,A}(A_2) = 34.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Prepare a compact fleet	**A2:** Prepare an expanded fleet
Statesofnature	**S1:** High rental demand	**S2:** Low rental demand
Events	Rental demand revealed after the fleet-size decision	
Consequences	Profit in thousand USD for each fleet–demand pair	

The only revenue component is a state-based variable component:

$$R(A_i, S_j) = q(A_i, S_j)r^S(S_j).$$

The only cost component is an alternative-based fixed component:

$$C(A_i, S_j) = F^{C,A}(A_i).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j)r^S(S_j) - F^{C,A}(A_i).$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)r^S(S_j), \\ C(A_i, S_j) &= F^{C,A}(A_i), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$6(9) = 54$	22	$54 - 22 = 32$
(A_1, S_2)	$3(6) = 18$	22	$18 - 22 = -4$
(A_2, S_1)	$8(9) = 72$	34	$72 - 34 = 38$
(A_2, S_2)	$4(6) = 24$	34	$24 - 34 = -10$

The resulting payoff table is:

Alternative	High rental demand	Low rental demand
Compact fleet	32	−4
Expanded fleet	38	−10

The negative payoffs in the low-demand state represent losses because the fixed preparation cost is greater than the revenue earned.

3.3 Community Arts Sponsorship (C1–C3)

A community arts organisation must select a sponsorship agreement before the event requirements are confirmed. Each agreement provides a fixed sponsorship payment, while the event incurs a fixed regulatory and safety cost determined by the state of nature.

The organisation is comparing two alternatives:

1. A_1: Accept the standard sponsorship agreement
2. A_2: Accept the premium sponsorship agreement

The possible states of nature are:

1. S_1: Routine permit requirements
2. S_2: Enhanced safety requirements

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 52, \qquad F^{R,A}(A_2) = 68.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 18, \qquad F^{C,S}(S_2) = 31.$$

Neither model contains a variable component, so quantity is not required.

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Accept the standard sponsorship agreement	**A2:** Accept the premium sponsorship agreement
Statesofnature	**S1:** Routine permit requirements	**S2:** Enhanced safety requirements
Events	Event requirements confirmed after the sponsorship decision	
Consequences	Profit in thousand USD for each sponsorship–requirement pair	

The only revenue component is an alternative-based fixed component:

$$R(A_i, S_j) = F^{R,A}(A_i).$$

The only cost component is a state-based fixed component:

$$C(A_i, S_j) = F^{C,S}(S_j).$$

Therefore,

$$P(A_i, S_j) = F^{R,A}(A_i) - F^{C,S}(S_j).$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$R(A_i, S_j) = F^{R,A}(A_i),$$
$$C(A_i, S_j) = F^{C,S}(S_j),$$
$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	52	18	$52 - 18 = 34$
(A_1, S_2)	52	31	$52 - 31 = 21$
(A_2, S_1)	68	18	$68 - 18 = 50$
(A_2, S_2)	68	31	$68 - 31 = 37$

The resulting payoff table is:

Alternative	Routine requirements	Enhanced requirements
Standard sponsorship	34	21
Premium sponsorship	50	37

3.4 Distribution Contract Fulfilment (C1–C3)

A distribution company must choose a fulfilment method before the volume of a new contract is known. The contract provides a fixed payment determined by the realised volume state, while the processing cost per unit depends on the selected fulfilment method.

The company is comparing two alternatives:

1. A_1: Use manual packing
2. A_2: Use automated packing

The possible states of nature are:

1. S_1: Standard-volume contract
2. S_2: High-volume contract

The quantities, measured in hundreds of processed orders, are

$$q(A_1, S_1) = 4, \quad q(A_1, S_2) = 7,$$
$$q(A_2, S_1) = 5, \quad q(A_2, S_2) = 8.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 70, \qquad F^{R,S}(S_2) = 104.$$

The alternative-based cost rates, in thousand USD per hundred orders, are

$$c^A(A_1) = 7, \qquad c^A(A_2) = 5.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use manual packing	**A2:** Use automated packing
Statesofnature	**S1:** Standard-volume contract	**S2:** High-volume contract
Events	Contract volume revealed after the fulfilment-method decision	
Consequences	Profit in thousand USD for each fulfilment-method–volume pair	

The only revenue component is a state-based fixed component:

$$R(A_i, S_j) = F^{R,S}(S_j).$$

The only cost component is an alternative-based variable component:

$$C(A_i, S_j) = q(A_i, S_j)c^A(A_i).$$

Therefore,

$$P(A_i, S_j) = F^{R,S}(S_j) - q(A_i, S_j)c^A(A_i).$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)c^A(A_i), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	70	$4(7) = 28$	$70 - 28 = 42$
(A_1, S_2)	104	$7(7) = 49$	$104 - 49 = 55$
(A_2, S_1)	70	$5(5) = 25$	$70 - 25 = 45$
(A_2, S_2)	104	$8(5) = 40$	$104 - 40 = 64$

The resulting payoff table is:

Alternative	Standard-volume contract	High-volume contract
Manual packing	42	55
Automated packing	45	64

4 Two-Component Models

Problems:

In a two-component model, revenue contains exactly two nonzero components and cost contains exactly two nonzero components. Each model is constructed independently, so the revenue and cost structures do not need to match.

The six paired patterns in this section are:

Problem	Revenue model	Cost model
9	$R(A_i, S_j) = q(A_i, S_j)\big[r^A(A_i) + r^S(S_j)\big]$	$C(A_i, S_j) = q(A_i, S_j)c^A(A_i) + F^{C,S}(S_j)$
10	$R(A_i, S_j) = q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i)$	$C(A_i, S_j) = q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i)$
11	$R(A_i, S_j) = q(A_i, S_j)r^A(A_i) + F^{R,S}(S_j)$	$C(A_i, S_j) = q(A_i, S_j)c^S(S_j) + F^{C,S}(S_j)$
12	$R(A_i, S_j) = q(A_i, S_j)r^S(S_j) + F^{R,A}(A_i)$	$C(A_i, S_j) = F^{C,A}(A_i) + F^{C,S}(S_j)$
13	$R(A_i, S_j) = q(A_i, S_j)r^S(S_j) + F^{R,S}(S_j)$	$C(A_i, S_j) = q(A_i, S_j)\big[c^A(A_i) + c^S(S_j)\big]$
14	$R(A_i, S_j) = F^{R,A}(A_i) + F^{R,S}(S_j)$	$C(A_i, S_j) = q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i)$

Thus, all six possible two-component structures—AV+SV, AV+AF, AV+SF, SV+AF, SV+SF, and AF+SF—appear exactly once as a revenue structure and exactly once as a cost structure.

4.1 Farm Produce Subscription Plans (C1–C3)

A farm cooperative must choose a produce-subscription plan before seasonal market conditions are known. Revenue per subscription consists of a rate determined by the selected plan and an additional rate determined by the market state. Costs consist of a plan-based variable cost and a fixed distribution cost determined by the state.

The cooperative is comparing two alternatives:

1. A_1: Offer a standard produce box
2. A_2: Offer a premium produce box

The possible states of nature are:

1. S_1: Strong seasonal demand
2. S_2: Weak seasonal demand

The quantities, measured in hundreds of subscriptions, are

$$q(A_1, S_1) = 7, \quad q(A_1, S_2) = 4,$$
$$q(A_2, S_1) = 6, \quad q(A_2, S_2) = 3.$$

The alternative-based revenue rates, in thousand USD per hundred subscriptions, are

$$r^A(A_1) = 8, \qquad r^A(A_2) = 11.$$

The state-based revenue rates are

$$r^S(S_1) = 3, \qquad r^S(S_2) = 1.$$

The alternative-based cost rates are

$$c^A(A_1) = 4, \qquad c^A(A_2) = 6.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 9, \qquad F^{C,S}(S_2) = 13.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Offer a standard produce box	**A2:** Offer a premium produce box
Statesofnature	**S1:** Strong seasonal demand	**S2:** Weak seasonal demand
Events	Seasonal market conditions revealed after the subscription-plan decision	
Consequences	Profit in thousand USD for each plan–market-state pair	

Revenue contains alternative-based and state-based variable components:

$$R(A_i, S_j) = q(A_i, S_j) \left[r^A(A_i) + r^S(S_j)\right].$$

Cost contains an alternative-based variable component and a state-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)c^A(A_i) + F^{C,S}(S_j).$$

Therefore,

$$P(A_i, S_j) = q(A_i, S_j) \left[r^A(A_i) + r^S(S_j)\right] - \left[q(A_i, S_j)c^A(A_i) + F^{C,S}(S_j)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j) \left[r^A(A_i) + r^S(S_j)\right], \\ C(A_i, S_j) &= q(A_i, S_j)c^A(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$7(8 + 3) = 77$	$7(4) + 9 = 37$	$77 - 37 = 40$
(A_1, S_2)	$4(8 + 1) = 36$	$4(4) + 13 = 29$	$36 - 29 = 7$
(A_2, S_1)	$6(11 + 3) = 84$	$6(6) + 9 = 45$	$84 - 45 = 39$
(A_2, S_2)	$3(11 + 1) = 36$	$3(6) + 13 = 31$	$36 - 31 = 5$

The resulting payoff table is:

Alternative	Strong seasonal demand	Weak seasonal demand
Standard produce box	40	7
Premium produce box	39	5

4.2 Mobile Catering Configuration (C1–C3)

A mobile-catering company must choose a service configuration before event demand and ingredient-market conditions are known. Revenue consists of a configuration-based rate per service order and a fixed booking payment determined by the selected configuration. Cost consists of a state-based rate per order and a fixed setup cost determined by the configuration.

The company is comparing two alternatives:

1. A_1: Prepare a compact catering station
2. A_2: Prepare a full mobile kitchen

The possible states of nature are:

1. S_1: Busy event with stable ingredient prices
2. S_2: Quiet event with higher ingredient prices

The quantities, measured in hundreds of service orders, are

$$q(A_1, S_1) = 8, \quad q(A_1, S_2) = 4,$$
$$q(A_2, S_1) = 10, \quad q(A_2, S_2) = 5.$$

The alternative-based revenue rates, in thousand USD per hundred orders, are

$$r^A(A_1) = 7, \qquad r^A(A_2) = 9.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 18, \qquad F^{R,A}(A_2) = 25.$$

The state-based cost rates, in thousand USD per hundred orders, are

$$c^S(S_1) = 3, \qquad c^S(S_2) = 5.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 20, \qquad F^{C,A}(A_2) = 32.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Prepare a compact catering station	**A2:** Prepare a full mobile kitchen
Statesofnature	**S1:** Busy event with stable ingredient prices	**S2:** Quiet event with higher ingredient prices
Events	Event demand and ingredient-market conditions revealed after the configuration decision	
Consequences	Profit in thousand USD for each configuration–market-state pair	

Revenue contains alternative-based variable and fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i).$$

Cost contains a state-based variable component and an alternative-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i).$$

Therefore,

$$P(A_i, S_j) = \left[q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i)\right] - \left[q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i), \\ C(A_i, S_j) &= q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$8(7) + 18 = 74$	$8(3) + 20 = 44$	$74 - 44 = 30$
(A_1, S_2)	$4(7) + 18 = 46$	$4(5) + 20 = 40$	$46 - 40 = 6$
(A_2, S_1)	$10(9) + 25 = 115$	$10(3) + 32 = 62$	$115 - 62 = 53$
(A_2, S_2)	$5(9) + 25 = 70$	$5(5) + 32 = 57$	$70 - 57 = 13$

The resulting payoff table is:

Alternative	Busy event	Quiet event
Compact catering station	30	6
Full mobile kitchen	53	13

4.3 Online Training Programme (C1–C3)

An online-training provider must select a programme before enrolment conditions are known. Revenue consists of a programme-based rate per enrolment group and a fixed payment determined by the enrolment state. Cost consists of a state-based delivery rate and a fixed platform cost determined by the state.

The provider is comparing two alternatives:

1. A_1: Offer the foundation programme
2. A_2: Offer the advanced programme

The possible states of nature are:

1. S_1: Corporate enrolment
2. S_2: Individual enrolment

The quantities, measured in enrolment groups, are

$$q(A_1, S_1) = 5, \quad q(A_1, S_2) = 3,$$
$$q(A_2, S_1) = 4, \quad q(A_2, S_2) = 2.$$

The alternative-based revenue rates, in thousand USD per group, are

$$r^A(A_1) = 14, \qquad r^A(A_2) = 20.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 12, \qquad F^{R,S}(S_2) = 7.$$

The state-based cost rates, in thousand USD per group, are

$$c^S(S_1) = 6, \qquad c^S(S_2) = 8.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 10, \qquad F^{C,S}(S_2) = 14.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Offer the foundation programme	**A2:** Offer the advanced programme
Statesofnature	**S1:** Corporate enrolment	**S2:** Individual enrolment
Events	Enrolment type revealed after the programme decision	
Consequences	Profit in thousand USD for each programme–enrolment pair	

Revenue contains an alternative-based variable component and a state-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)r^A(A_i) + F^{R,S}(S_j).$$

Cost contains state-based variable and fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c^S(S_j) + F^{C,S}(S_j).$$

Therefore,

$$P(A_i, S_j) = \left[q(A_i, S_j)r^A(A_i) + F^{R,S}(S_j)\right] - \left[q(A_i, S_j)c^S(S_j) + F^{C,S}(S_j)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned}
R(A_i, S_j) &= q(A_i, S_j)r^A(A_i) + F^{R,S}(S_j),\\
C(A_i, S_j) &= q(A_i, S_j)c^S(S_j) + F^{C,S}(S_j),\\
P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j).
\end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$5(14) + 12 = 82$	$5(6) + 10 = 40$	$82 - 40 = 42$
(A_1, S_2)	$3(14) + 7 = 49$	$3(8) + 14 = 38$	$49 - 38 = 11$
(A_2, S_1)	$4(20) + 12 = 92$	$4(6) + 10 = 34$	$92 - 34 = 58$
(A_2, S_2)	$2(20) + 7 = 47$	$2(8) + 14 = 30$	$47 - 30 = 17$

The resulting payoff table is:

Alternative	Corporate enrolment	Individual enrolment
Foundation programme	42	11
Advanced programme	58	17

4.4 Community Venue Membership (C1–C3)

A community organisation must select a venue-membership plan before booking conditions and permit requirements are known. Revenue consists of a state-based rate per booking and a fixed sponsorship payment determined by the selected membership plan. Costs consist of two fixed components: one determined by the plan and one determined by the state.

The organisation is comparing two alternatives:

1. A_1: Select the standard venue membership
2. A_2: Select the expanded venue membership

The possible states of nature are:

1. S_1: High booking demand
2. S_2: Low booking demand

The numbers of bookings are

$$q(A_1, S_1) = 6, \quad q(A_1, S_2) = 3,$$
$$q(A_2, S_1) = 8, \quad q(A_2, S_2) = 4.$$

The state-based revenue rates, in thousand USD per booking, are

$$r^S(S_1) = 10, \qquad r^S(S_2) = 7.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 15, \qquad F^{R,A}(A_2) = 22.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 28, \qquad F^{C,A}(A_2) = 40.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 8, \qquad F^{C,S}(S_2) = 13.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Select the standard venue membership	**A2:** Select the expanded venue membership
Statesofnature	**S1:** High booking demand	**S2:** Low booking demand
Events	Booking demand revealed after the membership-plan decision	
Consequences	Profit in thousand USD for each membership–demand pair	

Revenue contains a state-based variable component and an alternative-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i).$$

Cost contains alternative-based and state-based fixed components:

$$C(A_i, S_j) = F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$P(A_i, S_j) = \left[q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i)\right] - \left[F^{C,A}(A_i) + F^{C,S}(S_j)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$R(A_i, S_j) = q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i),$$
$$C(A_i, S_j) = F^{C,A}(A_i) + F^{C,S}(S_j),$$
$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i,S_j)$	$C(A_i,S_j)$	$P(A_i,S_j)=R(A_i,S_j)-C(A_i,S_j)$
(A_1,S_1)	$6(10)+15=75$	$28+8=36$	$75-36=39$
(A_1,S_2)	$3(7)+15=36$	$28+13=41$	$36-41=-5$
(A_2,S_1)	$8(10)+22=102$	$40+8=48$	$102-48=54$
(A_2,S_2)	$4(7)+22=50$	$40+13=53$	$50-53=-3$

The resulting payoff table is:

Alternative	High booking demand	Low booking demand
Standard venue membership	39	-5
Expanded venue membership	54	-3

The negative payoffs under low booking demand represent losses because the combined fixed costs exceed the revenue earned.

4.5 Urban Delivery Fleet (C1–C3)

An urban-delivery company must choose a fleet configuration before delivery demand and traffic conditions are known. Revenue consists of a state-based rate per delivery batch and a fixed performance payment determined by the state. Variable cost per batch combines a fleet-based operating rate and a state-based traffic surcharge.

The company is comparing two alternatives:

1. A_1: Use a conventional van fleet
2. A_2: Use an electric cargo-bike fleet

The possible states of nature are:

1. S_1: High demand with light traffic
2. S_2: Low demand with heavy traffic

The quantities, measured in delivery batches, are

$$q(A_1,S_1)=9, \quad q(A_1,S_2)=5,$$
$$q(A_2,S_1)=12, \quad q(A_2,S_2)=6.$$

The state-based revenue rates, in thousand USD per batch, are

$$r^S(S_1)=8, \qquad r^S(S_2)=6.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1)=14, \qquad F^{R,S}(S_2)=8.$$

The alternative-based cost rates, in thousand USD per batch, are

$$c^A(A_1)=3, \qquad c^A(A_2)=2.$$

The state-based cost rates, in thousand USD per batch, are

$$c^S(S_1)=1, \qquad c^S(S_2)=3.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use a conventional van fleet	**A2:** Use an electric cargo-bike fleet
Statesofnature	**S1:** High demand with light traffic	**S2:** Low demand with heavy traffic
Events	Delivery demand and traffic conditions revealed after the fleet decision	
Consequences	Profit in thousand USD for each fleet–market-state pair	

Revenue contains state-based variable and fixed components:

$$R(A_i, S_j) = q(A_i, S_j) r^S(S_j) + F^{R,S}(S_j).$$

Cost contains alternative-based and state-based variable components:

$$C(A_i, S_j) = q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right].$$

Therefore,

$$P(A_i, S_j) = \left[q(A_i, S_j) r^S(S_j) + F^{R,S}(S_j)\right] - q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$R(A_i, S_j) = q(A_i, S_j) r^S(S_j) + F^{R,S}(S_j),$$
$$C(A_i, S_j) = q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right],$$
$$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j).$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$9(8) + 14 = 86$	$9(3 + 1) = 36$	$86 - 36 = 50$
(A_1, S_2)	$5(6) + 8 = 38$	$5(3 + 3) = 30$	$38 - 30 = 8$
(A_2, S_1)	$12(8) + 14 = 110$	$12(2 + 1) = 36$	$110 - 36 = 74$
(A_2, S_2)	$6(6) + 8 = 44$	$6(2 + 3) = 30$	$44 - 30 = 14$

The resulting payoff table is:

Alternative	High demand, light traffic	Low demand, heavy traffic
Conventional van fleet	50	8
Electric cargo-bike fleet	74	14

4.6 Data-Service Contract (C1–C3)

A data-services company must choose a processing method before the requirements of a new contract are confirmed. Revenue consists of a fixed payment determined by the selected method and a fixed service-level payment determined by the contract state. Cost consists of a method-based processing rate and a fixed implementation cost determined by the selected method.

The company is comparing two alternatives:

1. A_1: Use standard data processing
2. A_2: Use automated data processing

The possible states of nature are:

1. S_1: Standard-volume contract
2. S_2: High-volume contract

The quantities, measured in processing batches, are

$$q(A_1, S_1) = 4, \quad q(A_1, S_2) = 7,$$
$$q(A_2, S_1) = 5, \quad q(A_2, S_2) = 9.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 55, \qquad F^{R,A}(A_2) = 72.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 16, \qquad F^{R,S}(S_2) = 9.$$

The alternative-based cost rates, in thousand USD per batch, are

$$c^A(A_1) = 6, \qquad c^A(A_2) = 4.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 18, \qquad F^{C,A}(A_2) = 31.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use standard data processing	**A2:** Use automated data processing
Statesofnature	**S1:** Standard-volume contract	**S2:** High-volume contract
Events	Contract requirements confirmed after the processing-method decision	
Consequences	Profit in thousand USD for each processing-method–contract-state pair	

Revenue contains alternative-based and state-based fixed components:

$$R(A_i, S_j) = F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost contains alternative-based variable and fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i).$$

Therefore,

$$P(A_i, S_j) = \left[F^{R,A}(A_i) + F^{R,S}(S_j)\right] - \left[q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i)\right].$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$55 + 16 = 71$	$4(6) + 18 = 42$	$71 - 42 = 29$
(A_1, S_2)	$55 + 9 = 64$	$7(6) + 18 = 60$	$64 - 60 = 4$
(A_2, S_1)	$72 + 16 = 88$	$5(4) + 31 = 51$	$88 - 51 = 37$
(A_2, S_2)	$72 + 9 = 81$	$9(4) + 31 = 67$	$81 - 67 = 14$

The resulting payoff table is:

Alternative	Standard-volume contract	High-volume contract
Standard data processing	29	4
Automated data processing	37	14

5 Three-Component Models

Problems:

In a three-component model, revenue contains exactly three nonzero components and cost contains exactly three nonzero components. Each model is constructed independently, so the revenue and cost structures do not need to match.

The four paired patterns in this section are:

Problem	Revenue model	Cost model
15	$R(A_i,S_j)=q(A_i,S_j)\big[r^A(A_i)+r^S(S_j)\big]+F^{R,A}(A_i)$	$C(A_i,S_j)=q(A_i,S_j)\big[c^A(A_i)+c^S(S_j)\big]+F^{C,S}(S_j)$
16	$R(A_i,S_j)=q(A_i,S_j)\big[r^A(A_i)+r^S(S_j)\big]+F^{R,S}(S_j)$	$C(A_i,S_j)=q(A_i,S_j)c^A(A_i)+F^{C,A}(A_i)+F^{C,S}(S_j)$
17	$R(A_i,S_j)=q(A_i,S_j)r^A(A_i)+F^{R,A}(A_i)+F^{R,S}(S_j)$	$C(A_i,S_j)=q(A_i,S_j)c^S(S_j)+F^{C,A}(A_i)+F^{C,S}(S_j)$
18	$R(A_i,S_j)=q(A_i,S_j)r^S(S_j)+F^{R,A}(A_i)+F^{R,S}(S_j)$	$C(A_i,S_j)=q(A_i,S_j)\big[c^A(A_i)+c^S(S_j)\big]+F^{C,A}(A_i)$

Thus, all four possible three-component structures—AV+SV+AF, AV+SV+SF, AV+AF+SF, and SV+AF+SF—appear exactly once as a revenue structure and exactly once as a cost structure.

5.1 Regional Parcel Delivery (C1–C3)

A regional logistics company must select a parcel-delivery plan before demand and traffic conditions are known. Revenue per delivery batch combines a rate determined by the selected plan and a rate determined by the state. The selected plan also provides a fixed service payment. Cost per batch combines a plan-based operating rate and a state-based traffic rate, together with a fixed distribution cost determined by the state.

The company is comparing two alternatives:

1. A_1: Use the standard delivery plan
2. A_2: Use the express delivery plan

The possible states of nature are:

1. S_1: High demand with light traffic
2. S_2: Moderate demand with heavy traffic

The quantities, measured in hundreds of delivery batches, are

$$q(A_1,S_1)=8,\quad q(A_1,S_2)=5,$$
$$q(A_2,S_1)=10,\quad q(A_2,S_2)=6.$$

The alternative-based revenue rates, in thousand USD per hundred delivery batches, are

$$r^A(A_1)=9,\qquad r^A(A_2)=11.$$

The state-based revenue rates are

$$r^S(S_1)=3,\qquad r^S(S_2)=2.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1)=16,\qquad F^{R,A}(A_2)=24.$$

The alternative-based cost rates are

$$c^A(A_1)=4,\qquad c^A(A_2)=5.$$

The state-based cost rates are

$$c^S(S_1)=1,\qquad c^S(S_2)=3.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1)=10,\qquad F^{C,S}(S_2)=15.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use the standard delivery plan	**A2:** Use the express delivery plan
Statesofnature	**S1:** High demand with light traffic	**S2:** Moderate demand with heavy traffic
Events	Demand and traffic conditions revealed after the delivery-plan decision	
Consequences	Profit in thousand USD for each delivery-plan–state pair	

Revenue contains alternative-based and state-based variable components and an alternative-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i).$$

Cost contains alternative-based and state-based variable components and a state-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = {} & \left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i)\right\} \\ & - \left\{q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i), \\ C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$8(9+3)+16=112$	$8(4+1)+10=50$	$112-50=62$
(A_1, S_2)	$5(9+2)+16=71$	$5(4+3)+15=50$	$71-50=21$
(A_2, S_1)	$10(11+3)+24=164$	$10(5+1)+10=70$	$164-70=94$
(A_2, S_2)	$6(11+2)+24=102$	$6(5+3)+15=63$	$102-63=39$

The resulting payoff table is:

Alternative	High demand, light traffic	Moderate demand, heavy traffic
Standard delivery plan	62	21
Express delivery plan	94	39

5.2 Solar Installation Programme (C1–C3)

A renewable-energy company must select a solar installation programme before customer demand and permit conditions are known. Revenue per installation batch combines a programme-based rate and a state-based market rate, together with a fixed incentive determined by the state. Costs consist of a programme-based variable cost, a fixed setup cost determined by the programme, and a fixed permit cost determined by the state.

The company is comparing two alternatives:

1. A_1: Offer the standard-panel programme
2. A_2: Offer the high-efficiency-panel programme

The possible states of nature are:

1. S_1: Strong commercial demand
2. S_2: Limited residential demand

The quantities, measured in installation batches, are

$$q(A_1, S_1) = 7, \quad q(A_1, S_2) = 4,$$
$$q(A_2, S_1) = 6, \quad q(A_2, S_2) = 3.$$

The alternative-based revenue rates, in thousand USD per installation batch, are

$$r^A(A_1) = 12, \qquad r^A(A_2) = 18.$$

The state-based revenue rates are

$$r^S(S_1) = 4, \qquad r^S(S_2) = 2.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 20, \qquad F^{R,S}(S_2) = 11.$$

The alternative-based cost rates, in thousand USD per installation batch, are

$$c^A(A_1) = 6, \qquad c^A(A_2) = 8.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 18, \qquad F^{C,A}(A_2) = 25.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 9, \qquad F^{C,S}(S_2) = 14.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Offer the standard-panel programme	**A2:** Offer the high-efficiency-panel programme
Statesofnature	**S1:** Strong commercial demand	**S2:** Limited residential demand
Events	Customer demand and permit conditions revealed after the programme decision	
Consequences	Profit in thousand USD for each programme–market-state pair	

Revenue contains alternative-based and state-based variable components and a state-based fixed component:

$$R(A_i, S_j) = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,S}(S_j).$$

Cost contains an alternative-based variable component and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = \; & \left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,S}(S_j)\right\} \\ & - \left\{q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)c^A(A_i) + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$7(12 + 4) + 20 = 132$	$7(6) + 18 + 9 = 69$	$132 - 69 = 63$
(A_1, S_2)	$4(12 + 2) + 11 = 67$	$4(6) + 18 + 14 = 56$	$67 - 56 = 11$
(A_2, S_1)	$6(18 + 4) + 20 = 152$	$6(8) + 25 + 9 = 82$	$152 - 82 = 70$
(A_2, S_2)	$3(18 + 2) + 11 = 71$	$3(8) + 25 + 14 = 63$	$71 - 63 = 8$

The resulting payoff table is:

Alternative	Strong commercial demand	Limited residential demand
Standard-panel programme	63	11
High-efficiency-panel programme	70	8

5.3 Hybrid Conference Production (C1–C3)

An events company must choose a hybrid-conference production package before attendance and sponsorship conditions are known. Revenue consists of a package-based rate per event segment, a fixed booking payment determined by the selected package, and a fixed sponsorship payment determined by the state. Costs consist of a state-based delivery rate, a fixed setup cost determined by the package, and a fixed licensing cost determined by the state.

The company is comparing two alternatives:

1. A_1: Use the standard hybrid package
2. A_2: Use the immersive hybrid package

The possible states of nature are:

1. S_1: Strong attendance and sponsorship
2. S_2: Limited attendance and sponsorship

The quantities, measured in event segments, are

$$q(A_1, S_1) = 6, \quad q(A_1, S_2) = 4,$$
$$q(A_2, S_1) = 8, \quad q(A_2, S_2) = 5.$$

The alternative-based revenue rates, in thousand USD per event segment, are

$$r^A(A_1) = 13, \qquad r^A(A_2) = 16.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 14, \qquad F^{R,A}(A_2) = 22.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 18, \qquad F^{R,S}(S_2) = 9.$$

The state-based cost rates, in thousand USD per event segment, are

$$c^S(S_1) = 5, \qquad c^S(S_2) = 7.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 24, \qquad F^{C,A}(A_2) = 37.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 8, \qquad F^{C,S}(S_2) = 13.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use the standard hybrid package	**A2:** Use the immersive hybrid package
Statesofnature	**S1:** Strong attendance and sponsorship	**S2:** Limited attendance and sponsorship
Events	Attendance and sponsorship conditions revealed after the package decision	
Consequences	Profit in thousand USD for each production-package–state pair	

Revenue contains an alternative-based variable component and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost contains a state-based variable component and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = \; & \left\{q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i) + F^{R,S}(S_j)\right\} \\ & - \left\{q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)r^A(A_i) + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)c^S(S_j) + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$6(13) + 14 + 18 = 110$	$6(5) + 24 + 8 = 62$	$110 - 62 = 48$
(A_1, S_2)	$4(13) + 14 + 9 = 75$	$4(7) + 24 + 13 = 65$	$75 - 65 = 10$
(A_2, S_1)	$8(16) + 22 + 18 = 168$	$8(5) + 37 + 8 = 85$	$168 - 85 = 83$
(A_2, S_2)	$5(16) + 22 + 9 = 111$	$5(7) + 37 + 13 = 85$	$111 - 85 = 26$

The resulting payoff table is:

Alternative	Strong attendance and sponsorship	Limited attendance and sponsorship
Standard hybrid package	48	10
Immersive hybrid package	83	26

5.4 Cold-Storage Export Contract (C1–C3)

A food-export company must select a cold-storage configuration before export demand and energy-market conditions are known. Revenue consists of a state-based rate per storage batch, a fixed capacity payment determined by the selected configuration, and a fixed export payment determined by the state. Variable cost per batch combines a configuration-based operating rate and a state-based energy surcharge. The selected configuration also produces a fixed setup cost.

The company is comparing two alternatives:

1. A_1: Use the modular cold-storage facility
2. A_2: Use the large cold-storage facility

The possible states of nature are:

1. S_1: Strong export demand
2. S_2: Limited export demand

The quantities, measured in storage batches, are

$$q(A_1, S_1) = 9, \quad q(A_1, S_2) = 5,$$
$$q(A_2, S_1) = 12, \quad q(A_2, S_2) = 7.$$

The state-based revenue rates, in thousand USD per storage batch, are

$$r^S(S_1) = 10, \qquad r^S(S_2) = 7.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 20, \qquad F^{R,A}(A_2) = 30.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 16, \qquad F^{R,S}(S_2) = 8.$$

The alternative-based cost rates, in thousand USD per storage batch, are

$$c^A(A_1) = 4, \qquad c^A(A_2) = 3.$$

The state-based cost rates are

$$c^S(S_1) = 2, \qquad c^S(S_2) = 4.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 28, \qquad F^{C,A}(A_2) = 42.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use the modular cold-storage facility	**A2:** Use the large cold-storage facility
Statesofnature	**S1:** Strong export demand	**S2:** Limited export demand
Events	Export demand and energy-market conditions revealed after the facility decision	
Consequences	Profit in thousand USD for each storage-facility–state pair	

Revenue contains a state-based variable component and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost contains alternative-based and state-based variable components and an alternative-based fixed component:

$$C(A_i, S_j) = q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = &\ \left\{q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i) + F^{R,S}(S_j)\right\} \\ &- \left\{q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j) r^S(S_j) + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j) \left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$9(10) + 20 + 16 = 126$	$9(4 + 2) + 28 = 82$	$126 - 82 = 44$
(A_1, S_2)	$5(7) + 20 + 8 = 63$	$5(4 + 4) + 28 = 68$	$63 - 68 = -5$
(A_2, S_1)	$12(10) + 30 + 16 = 166$	$12(3 + 2) + 42 = 102$	$166 - 102 = 64$
(A_2, S_2)	$7(7) + 30 + 8 = 87$	$7(3 + 4) + 42 = 91$	$87 - 91 = -4$

The resulting payoff table is:

Alternative	Strong export demand	Limited export demand
Modular cold-storage facility	44	-5
Large cold-storage facility	64	-4

The negative payoffs under limited export demand represent losses because the revenue earned does not cover the variable and fixed costs.

6 Complete Four-Component Models

Problems:

In a complete four-component model, both revenue and cost contain all four possible components:

1. an alternative-based variable component;
2. a state-based variable component;
3. an alternative-based fixed component;
4. a state-based fixed component.

Therefore, the complete revenue and cost models are

$$
\begin{aligned}
R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j),\\
C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j).
\end{aligned}
$$

The payoff for each alternative–state pair is consequently

$$
\begin{aligned}
P(A_i, S_j) = {} & \left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j)\right\}\\
& - \left\{q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}.
\end{aligned}
$$

The three problems in this section use 2×2, 3×2, and 2×3 alternative–state structures, respectively.

6.1 Smart Manufacturing Contract (C1–C3)

A manufacturing company must select a production system before product demand and input-market conditions are known. Revenue per production batch combines a rate determined by the selected system and a rate determined by the state. Revenue also includes a fixed contract payment determined by the system and a fixed market incentive determined by the state.

Variable cost per batch combines a system-based operating rate and a state-based material surcharge. Fixed cost consists of a setup cost determined by the selected system and a compliance cost determined by the state.

The company is comparing two alternatives:

1. A_1: Use the conventional assembly system
2. A_2: Use the automated assembly system

The possible states of nature are:

1. S_1: Strong demand with favourable input conditions
2. S_2: Limited demand with expensive inputs

The quantities, measured in production batches, are

$$
\begin{aligned}
q(A_1, S_1) &= 8, \quad q(A_1, S_2) = 5,\\
q(A_2, S_1) &= 11, \quad q(A_2, S_2) = 6.
\end{aligned}
$$

The alternative-based revenue rates, in thousand USD per production batch, are

$$r^A(A_1) = 12, \qquad r^A(A_2) = 15.$$

The state-based revenue rates, in thousand USD per production batch, are

$$r^S(S_1) = 4, \qquad r^S(S_2) = 2.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 18, \qquad F^{R,A}(A_2) = 28.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 14, \qquad F^{R,S}(S_2) = 7.$$

The alternative-based cost rates, in thousand USD per production batch, are

$$c^A(A_1) = 5, \qquad c^A(A_2) = 6.$$

The state-based cost rates, in thousand USD per production batch, are

$$c^S(S_1) = 2, \qquad c^S(S_2) = 4.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 24, \qquad F^{C,A}(A_2) = 38.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 9, \qquad F^{C,S}(S_2) = 15.$$

Questions/Tasks.

- **C1** Interpreting decision alternatives, events, consequences, and states.
- **C3** Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description	
Alternatives	**A1:** Use the conventional assembly system	**A2:** Use the automated assembly system
Statesofnature	**S1:** Strong demand with favourable input conditions	**S2:** Limited demand with expensive inputs
Events	Demand and input-market conditions revealed after the production-system decision	
Consequences	Profit in thousand USD for each production-system–state pair	

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = &\left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j)\right\} \\ &- \left\{q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$8(12+4)+18+14=160$	$8(5+2)+24+9=89$	$160-89=71$
(A_1, S_2)	$5(12+2)+18+7=95$	$5(5+4)+24+15=84$	$95-84=11$
(A_2, S_1)	$11(15+4)+28+14=251$	$11(6+2)+38+9=135$	$251-135=116$
(A_2, S_2)	$6(15+2)+28+7=137$	$6(6+4)+38+15=113$	$137-113=24$

The resulting payoff table is:

Alternative	Strong demand, favourable inputs	Limited demand, expensive inputs
Conventional assembly system	71	11
Automated assembly system	116	24

6.2 Distribution Centre Automation (C1–C3)

A retail company must select a distribution-centre configuration before order volume and energy-market conditions are known. Revenue per order batch combines a rate determined by the selected configuration and a rate determined by the state. Revenue also includes a fixed service payment determined by the configuration and a fixed delivery incentive determined by the state.

Variable cost per order batch combines a configuration-based operating rate and a state-based energy surcharge. Fixed cost consists of an installation cost determined by the configuration and a compliance cost determined by the state.

The company is comparing three alternatives:

1. A_1: Use the manual distribution configuration
2. A_2: Use the semi-automated distribution configuration

3. A_3: Use the robotic distribution configuration

The possible states of nature are:

1. S_1: High order volume with favourable energy prices

2. S_2: Moderate order volume with expensive energy

The quantities, measured in hundreds of order batches, are

$$\begin{aligned} q(A_1, S_1) &= 8, \quad & q(A_1, S_2) &= 5, \\ q(A_2, S_1) &= 11, \quad & q(A_2, S_2) &= 7, \\ q(A_3, S_1) &= 14, \quad & q(A_3, S_2) &= 8. \end{aligned}$$

The alternative-based revenue rates, in thousand USD per hundred order batches, are

$$r^A(A_1) = 12, \quad r^A(A_2) = 15, \quad r^A(A_3) = 18.$$

The state-based revenue rates are

$$r^S(S_1) = 4, \qquad r^S(S_2) = 2.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 16, \quad F^{R,A}(A_2) = 24, \quad F^{R,A}(A_3) = 35.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 15, \qquad F^{R,S}(S_2) = 8.$$

The alternative-based cost rates, in thousand USD per hundred order batches, are

$$c^A(A_1) = 5, \quad c^A(A_2) = 6, \quad c^A(A_3) = 8.$$

The state-based cost rates are

$$c^S(S_1) = 2, \qquad c^S(S_2) = 4.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 22, \quad F^{C,A}(A_2) = 34, \quad F^{C,A}(A_3) = 50.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 9, \qquad F^{C,S}(S_2) = 16.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Use the manual distribution configuration	**A2:** Use the semi-automated distribution configuration	**A3:** Use the robotic distribution configuration
Statesofnature	**S1:** High order volume with favourable energy prices	**S2:** Moderate order volume with expensive energy	
Events	Order volume and energy-market conditions revealed after the distribution-configuration decision		
Consequences	Profit in thousand USD for each distribution-configuration–state pair		

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = &\left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j)\right\} \\ &- \left\{q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$8(12+4)+16+15=159$	$8(5+2)+22+9=87$	$159-87=72$
(A_1, S_2)	$5(12+2)+16+8=94$	$5(5+4)+22+16=83$	$94-83=11$
(A_2, S_1)	$11(15+4)+24+15=248$	$11(6+2)+34+9=131$	$248-131=117$
(A_2, S_2)	$7(15+2)+24+8=151$	$7(6+4)+34+16=120$	$151-120=31$
(A_3, S_1)	$14(18+4)+35+15=358$	$14(8+2)+50+9=199$	$358-199=159$
(A_3, S_2)	$8(18+2)+35+8=203$	$8(8+4)+50+16=162$	$203-162=41$

The resulting payoff table is:

Alternative	High order volume	Moderate order volume
Manual distribution configuration	72	11
Semi-automated distribution configuration	117	31
Robotic distribution configuration	159	41

6.3 Electric Vehicle Charging Network (C1–C3)

An energy company must select an electric vehicle charging-network design before charging demand and electricity-market conditions are known. Revenue per charging-session batch combines a rate determined by the selected network and a rate determined by the state. Revenue also includes a fixed service payment determined by the network and a fixed utilisation incentive determined by the state.

Variable cost per charging-session batch combines a network-based equipment rate and a state-based electricity rate. Fixed cost consists of an infrastructure cost determined by the network and a maintenance cost determined by the state.

The company is comparing two alternatives:

1. A_1: Build the neighbourhood charging network
2. A_2: Build the rapid-charging network

The possible states of nature are:

1. S_1: High utilisation with favourable electricity prices
2. S_2: Moderate utilisation with standard electricity prices
3. S_3: Low utilisation with expensive electricity

The quantities, measured in hundreds of charging-session batches, are

$$q(A_1, S_1) = 10, \quad q(A_1, S_2) = 7, \quad q(A_1, S_3) = 4,$$
$$q(A_2, S_1) = 14, \quad q(A_2, S_2) = 9, \quad q(A_2, S_3) = 5.$$

The alternative-based revenue rates, in thousand USD per hundred charging-session batches, are

$$r^A(A_1) = 11, \qquad r^A(A_2) = 16.$$

The state-based revenue rates are

$$r^S(S_1) = 4, \quad r^S(S_2) = 3, \quad r^S(S_3) = 2.$$

The alternative-based fixed revenues, in thousand USD, are

$$F^{R,A}(A_1) = 18, \qquad F^{R,A}(A_2) = 30.$$

The state-based fixed revenues, in thousand USD, are

$$F^{R,S}(S_1) = 16, \quad F^{R,S}(S_2) = 10, \quad F^{R,S}(S_3) = 5.$$

The alternative-based cost rates, in thousand USD per hundred charging-session batches, are

$$c^A(A_1) = 4, \qquad c^A(A_2) = 7.$$

The state-based cost rates are

$$c^S(S_1) = 2, \quad c^S(S_2) = 3, \quad c^S(S_3) = 5.$$

The alternative-based fixed costs, in thousand USD, are

$$F^{C,A}(A_1) = 25, \qquad F^{C,A}(A_2) = 45.$$

The state-based fixed costs, in thousand USD, are

$$F^{C,S}(S_1) = 8, \quad F^{C,S}(S_2) = 12, \quad F^{C,S}(S_3) = 18.$$

Questions/Tasks.

C1 Interpreting decision alternatives, events, consequences, and states.

C3 Calculating profit and building the payoff table.

C1 - Interpreting decision alternatives, events, consequences, and states.

Element	Description		
Alternatives	**A1:** Build the neighbourhood charging network	**A2:** Build the rapid-charging network	
Statesofnature	**S1:** High utilisation with favourable electricity prices	**S2:** Moderate utilisation with standard electricity prices	**S3:** Low utilisation with expensive electricity
Events	Charging demand and electricity-market conditions revealed after the network-design decision		
Consequences	Profit in thousand USD for each charging-network–state pair		

Revenue contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$R(A_i, S_j) = q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j).$$

Cost also contains alternative-based and state-based variable components and alternative-based and state-based fixed components:

$$C(A_i, S_j) = q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j).$$

Therefore,

$$\begin{aligned} P(A_i, S_j) = &\left\{q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j)\right\} \\ &- \left\{q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j)\right\}. \end{aligned}$$

C3 - Calculating profit and building the payoff table.

For every alternative–state pair, calculate

$$\begin{aligned} R(A_i, S_j) &= q(A_i, S_j)\left[r^A(A_i) + r^S(S_j)\right] + F^{R,A}(A_i) + F^{R,S}(S_j), \\ C(A_i, S_j) &= q(A_i, S_j)\left[c^A(A_i) + c^S(S_j)\right] + F^{C,A}(A_i) + F^{C,S}(S_j), \\ P(A_i, S_j) &= R(A_i, S_j) - C(A_i, S_j). \end{aligned}$$

Cell	$R(A_i, S_j)$	$C(A_i, S_j)$	$P(A_i, S_j) = R(A_i, S_j) - C(A_i, S_j)$
(A_1, S_1)	$10(11+4)+18+16=184$	$10(4+2)+25+8=93$	$184-93=91$
(A_1, S_2)	$7(11+3)+18+10=126$	$7(4+3)+25+12=86$	$126-86=40$
(A_1, S_3)	$4(11+2)+18+5=75$	$4(4+5)+25+18=79$	$75-79=-4$
(A_2, S_1)	$14(16+4)+30+16=326$	$14(7+2)+45+8=179$	$326-179=147$
(A_2, S_2)	$9(16+3)+30+10=211$	$9(7+3)+45+12=147$	$211-147=64$
(A_2, S_3)	$5(16+2)+30+5=125$	$5(7+5)+45+18=123$	$125-123=2$

The resulting payoff table is:

Alternative	High utilisation	Moderate utilisation	Low utilisation
Neighbourhood charging network	91	40	−4
Rapid-charging network	147	64	2

The negative payoff for the neighbourhood charging network under low utilisation represents a loss because the revenue earned does not cover the variable and fixed costs.